

IFZA IFZA

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Introduction

I am a Master's student in Computer Science at RPTU Kaiserslautern, Germany, specializing in Intelligence Systems and Software Engineering. As a Research Assistant at DFKI Kaiserslautern, I work on computer vision projects, including transformer-based object detection and semi-supervised learning techniques, for applications in document layout analysis and medical imaging.

Education

RPTU Kaiserslautern, Germany

Master of Science (M.Sc.) in Computer Science

Oct 2023 – Present

Kaiserslautern, Germany

- German Grade: 2.2/4.0
- Relevant Courses: Machine Learning I, Very Deep Learning, Generative AI Engineering, Practical Machine Learning & NLP with AWS, Image Generation and Enhancement, Data Visualization.

University of Punjab, Gujranwala, Pakistan

Bachelor of Science in Information Technology

Sep 2018 – Jun 2022

Gujranwala, Pakistan

- German Grade: 1.3/4.0

Technical Skills

Programming Languages: Python, C#, C++, JavaScript, HTML, CSS

Tools and Technologies: MS Visual Studio, Android Studio, NetSuite Oracle, MS SQL

Soft Skills: Team Player, Communication, Creativity, Self Learning, Critical Thinking

Work Experience

Research Assistant (Hiwi), DFKI Kaiserslautern

Research Assistant

Apr 2024 – Present

Kaiserslautern, Germany

- Designed and developed end-to-end computer vision pipelines for document layout analysis and medical image anomaly detection within the LUMINOUS, AIRISE, and SHARESPACE research projects.
- Implemented and evaluated transformer-based object detection and semi-supervised learning methods, focusing on query efficiency, robustness, and low-label regimes.
- Conducted experimental design, and quantitative evaluations, contributing to multiple peer-reviewed publications

Operations Analytics Specialist, DIGISQUAD

Operations Analytics and Systems Role

Apr 2021 – Oct 2023

Gujranwala, Pakistan

- Designed and maintained data-driven workflows for inventory management, order fulfillment, and market acquisition to support system-level decision support and process optimization.
- Developed and optimized analytical pipelines for product, sales, and competitor data using market analytics platforms as Helium10, enabling scalable monitoring and demand estimation.
- Automated reporting, performance tracking, and issue analysis processes, improving system reliability, reducing manual overhead, and increasing operational efficiency.

Academic Projects

Semi-Supervised Document Layout Analysis with Focused Attention and Denoising

Master's Project

- Contributed to the development of an end-to-end semi-supervised document layout analysis framework using focused attention and guided query mechanisms for efficient detection.
- Designed a hybrid, adaptive query strategy to improve generalization while reducing training cost.
- Evaluated on PubLayNet, PubTables, and TableBank, achieving up to 96.2% mAP and 98.3% recall on PubLayNet with only 10% labeled data.

Efficient Additive Attention for Semi-Supervised Document Layout Analysis

Master's Project

- Contributed to a semi-supervised framework integrating an efficient additive attention mechanism to reduce self-attention complexity while preserving global context.
- Evaluated robustness under input perturbations, demonstrating stable performance across noisy and distorted document layouts.

- Evaluated on PubLayNet, DocLayNet, and D4LA, achieving 96.8% mAP with 10% labels, outperforming Deformable-semi and SAM-semi while reducing training to 12 epochs.

FD-SSD: Semi-Supervised Detection of Bone Fenestration and Dehiscence

Master's Project

- Contributed to a semi-supervised framework for detecting bone fenestration and dehiscence in intraoral images, addressing the limitations of costly CBCT-based diagnosis.
- Designed a architecture with adaptive query generation from pseudo-labels, multi-scale feature learning, and query filtering to improve detection robustness.
- Leveraged both labeled and unlabeled data to reduce dependency on large annotated datasets and pre-training while improving generalization.
- Outperformed supervised baselines, improving mAP from 62.08% to 68.3% over DINO and from 65.67% to 68.3% over FD-SOS.

LLM-Based Multi-Agent System for Automated Literature Review

Master's Project

- Designed a multi-agent system leveraging the LLM Gemma2-9B-IT and the Groq API to automatically download, summarize, and generate literature reviews from research papers.
- Implemented specialized agents for paper extraction, reading, summarization, review writing, and human-in-the-loop.
- Evaluated generated reviews against manually written reviews using cosine similarity and ROUGE metrics, demonstrating moderate semantic similarity with low content overlap.
- Implemented round-robin agent communication with termination conditions to ensure efficient and stable multi-agent collaboration.

Seminars

Adversarial Attacks to Counterfactual Explanations

Master's Seminar

- Surveyed adversarial approaches for counterfactual explanations across multiple data modalities, focusing on interpretability, robustness, scalability, and ethical challenges in trustworthy AI.

Publications

Tahira Shehzadi, **Ifza Ifza**, Didier Stricker, Muhammad Zeshan Afzal, "FD-SSD: Semi-supervised Detection of Bone Fenestration and Dehiscence in Intraoral Images", *MIUA 2025*, pp. 3–14.

Tahira Shehzadi, **Ifza Ifza**, Didier Stricker, Muhammad Zeshan Afzal, "DocSemi: Efficient Document Layout Analysis with Guided Queries", *ICCV Workshops 2025*, pp. 7536-7546.

Tahira Shehzadi, **Ifza Ifza**, Didier Stricker, Muhammad Zeshan Afzal, "Efficient Additive Attention for Transformer-based Semi-supervised Document Layout Analysis", *ICCV Workshops 2025*, pp. 3495-3503.

Tahira Shehzadi, **Ifza Ifza**, Marcus Liwicki, Didier Stricker, Muhammad Zeshan Afzal, "Semi-Supervised Object Detection: A Survey on Progress from CNN to Transformer", *Sensors*, vol. 26, no. 1, p. 310, 2026.

Tahira Shehzadi, Rabya Noor, **Ifza Ifza**, Marcus Liwicki, Didier Stricker, Muhammad Zeshan Afzal, "Knowledge Distillation in Object Detection: A Survey from CNN to Transformer", *Sensors*, vol. 26, no. 1, p. 292, 2026.

Certifications

AWS Academy Graduate – Machine Learning Foundations (Jun 2025)

AWS Academy Graduate – Machine Learning for Natural Language Processing (Oct 2025)

Interests

Calligraphy, Art and Craft, Traveling